

# A COUNTEREXAMPLE TO HE AND HUANG'S RUNGE–KUTTA NECESSITY CONJECTURE

ABSTRACT. He and Huang conjectured that the sufficient  $Q$ - and  $D$ -condition package in their Theorem 1.2 is also necessary for Runge–Kutta order. We disprove the conjecture under the printed Hadamard condition

$$QO(m) : b \odot Q_m = \{0\}.$$

For every internally consistent explicit tableau and every admissible order  $p \geq 5$ , this condition forces  $b_2 c_2^2 = 0$ . The embedded formula in Verner's pair RK(8–6:5) $a$  has order exactly five and satisfies

$$\widehat{b}_2 = -\frac{1}{5}, \quad c_2 = \frac{1}{5},$$

so the proposed necessary package fails for every admissible pair  $(m, n)$ . The argument concerns the printed Hadamard product and does not apply to scalar orthogonality.

## 1. THE OBSTRUCTION

He and Huang define, for nonnegative integers  $r$ ,

$$q_r = A c^{\odot r} - \frac{1}{r+1} c^{\odot(r+1)},$$

and

$$Q_1 = \text{span}\{q_0\}, \quad Q_2 = \text{span}\{q_1, A q_0, q_0 \odot c, q_0\}.$$

Their spaces satisfy

$$Q_1 \subseteq Q_2 \subseteq Q_3 \subseteq \cdots$$

[1, Defs. 3.4 and 3.9, Eq. (16)]. The two entries in the bibliography below share the arXiv identifier 2605.16995: that record was retitled and reduced to a technical appendix at v3, and v3 carries no definitions or notation of its own, recalling the general theory instead from the v1-titled work, which it cites under that title. We therefore cite definitions and notation to v1, and Theorem 1.2 and Conjecture 2.1 to v3.

Theorem 1.2 of arXiv:2605.16995v3 states that an explicit Runge–Kutta method has order at least  $p$  if, for some nonnegative integers  $m, n$  with

$$m \geq n - 1, \quad m + n + 1 \geq p,$$

six conditions hold: the quadrature conditions  $B(p)$ , the  $Q$ -orthogonality conditions

$$QO(m) : \quad b \odot Q_m = \{0\},$$

the  $D$ -orthogonality conditions  $DO(n)$ , the weak mutual orthogonality conditions  $QD_{\text{weak}}(m, n)$ , the pivot residual condition  $PR(n)$ , and the  $Q$ -ring condition  $QR(m)$  [2, Thm. 1.2]. Conjecture 2.1 reads, verbatim [2, Conj. 2.1]:

The set of conditions in [Theorem 1.2] is not only sufficient [sic] but also necessary for a vector-valued explicit/implicit RK method to achieve order  $p$ .

Here  $\odot$  is the componentwise Hadamard product; the same source uses  $\cdot$  for the scalar inner product [1, Notations 2.1 and 2.3]. The Hadamard reading of  $QO(m)$  is forced by the source rather than inferred: all set operations there are declared to be Minkowski operations, so  $b \odot Q_m = \{0\}$  abbreviates  $\{b \odot q : q \in Q_m\} = \{0\}$ , that is,  $b \odot q = 0$  as a vector for every  $q \in Q_m$  [1, Notation 2.5]. The choice of product is deliberate within Theorem 1.2 itself, whose conditions  $DO(n)$  and  $QD_{\text{weak}}(m, n)$  are written with  $\cdot$  and not with  $\odot$ .

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Necessity of the package asserts that every one of the six conditions holds whenever the method has order  $p$ , so the conjecture is refuted by a single method of order  $p$  that violates a single condition for every admissible  $(m, n)$ . We exhibit one violating  $QO(m)$ .

**Proposition 1.** *Let  $(A, b, c)$  be an internally consistent explicit Runge-Kutta tableau, so that  $A$  is strictly lower triangular and  $c = A\mathbf{1}$ . Let  $p \geq 5$ . If nonnegative integers  $m, n$  satisfy*

$$m \geq n - 1, \quad m + n + 1 \geq p,$$

*and  $QO(m)$  holds, then*

$$b_2 c_2^2 = 0.$$

*Proof.* Since  $A$  is strictly lower triangular its first row vanishes, so internal consistency gives  $c_1 = (A\mathbf{1})_1 = 0$ . The vector  $q_1$  is one of the listed generators of  $Q_2$ , so in particular  $q_1 \in Q_2$ . The second row of  $A$  has  $a_{21}$  as its only possibly nonzero entry, so

$$(q_1)_2 = (Ac)_2 - \frac{1}{2}c_2^2 = a_{21}c_1 - \frac{1}{2}c_2^2 = -\frac{1}{2}c_2^2.$$

Moreover,  $m \geq n - 1$  implies  $n \leq m + 1$ , whence

$$p \leq m + n + 1 \leq 2m + 2.$$

Because  $p \geq 5$ , one has  $m \geq 2$ . Thus

$$q_1 \in Q_2 \subseteq Q_m,$$

and  $QO(m)$  gives

$$0 = (b \odot q_1)_2 = -\frac{1}{2}b_2 c_2^2.$$

□

## 2. AN ORDER-FIVE COUNTEREXAMPLE

**Theorem 2.** *Under the printed Hadamard definition of  $QO(m)$ , Conjecture 2.1 in [arXiv:2605.16995v3](#) is false. It already fails for an explicit Runge-Kutta method of order exactly five.*

*Proof.* Consider the embedded formula in Verner's eight-stage pair RK(8–6:5) $a$  [3, Table 4 and Algorithm 1.2]:

|               |                      |                      |                        |                        |                         |                    |                   |               |  |
|---------------|----------------------|----------------------|------------------------|------------------------|-------------------------|--------------------|-------------------|---------------|--|
| 0             |                      |                      |                        |                        |                         |                    |                   |               |  |
| $\frac{1}{5}$ | $\frac{1}{5}$        |                      |                        |                        |                         |                    |                   |               |  |
| $\frac{1}{4}$ | $\frac{3}{32}$       | $\frac{5}{32}$       |                        |                        |                         |                    |                   |               |  |
| $\frac{4}{7}$ | $\frac{60}{343}$     | $-\frac{440}{343}$   | $\frac{576}{343}$      |                        |                         |                    |                   |               |  |
| $\frac{7}{9}$ | $\frac{847}{17496}$  | $\frac{770}{729}$    | $-\frac{17024}{19683}$ | $\frac{84721}{157464}$ |                         |                    |                   |               |  |
| $\frac{1}{5}$ | $\frac{523}{2240}$   | 0                    | $-\frac{5}{57}$        | $\frac{245}{2496}$     | $-\frac{1215}{27664}$   |                    |                   |               |  |
| 1             | $\frac{15}{2212}$    | $\frac{314}{395}$    | $\frac{3448}{1501}$    | $-\frac{2695}{4108}$   | $\frac{203391}{273182}$ | $-\frac{864}{395}$ |                   |               |  |
| $\frac{4}{7}$ | $\frac{1185}{10976}$ | $-\frac{2543}{1715}$ | $\frac{36262}{19551}$  | $-\frac{245}{1248}$    | $\frac{1215}{13832}$    | $\frac{1}{5}$      | 0                 |               |  |
|               | $\frac{43}{560}$     | $-\frac{1}{5}$       | $\frac{2816}{7695}$    | $-\frac{41}{84240}$    | $\frac{19683}{69160}$   | $\frac{1}{5}$      | $\frac{79}{1080}$ | $\frac{1}{5}$ |  |

Let  $\widehat{b}$  denote the displayed weight vector. The order conditions are indexed by rooted trees: with elementary weights  $\Phi(\bullet) = \mathbf{1}$  and  $\Phi([t_1, \dots, t_k])_i = \prod_{j=1}^k (A\Phi(t_j))_i$ , and densities  $\gamma(\bullet) = 1$  and  $\gamma(t) = |t| \prod_{j=1}^k \gamma(t_j)$ , the method has order  $p$  precisely when  $\widehat{b}^\top \Phi(t) = 1/\gamma(t)$  for every rooted tree  $t$  with  $|t| \leq p$ . There are 1, 1, 2, 4, 9 such trees at orders one through five, and all seventeen conditions hold exactly on the displayed tableau; of the twenty trees of order six, eight give conditions that fail. For instance, the chain conditions of orders three, four and five hold,

$$\widehat{b}^\top Ac = \frac{1}{6}, \quad \widehat{b}^\top A^2c = \frac{1}{24}, \quad \widehat{b}^\top A^3c = \frac{1}{120},$$

while the one of order six fails:

$$\widehat{b}^\top A^4c = \frac{1}{672} \neq \frac{1}{720}.$$

Thus the formula has order exactly five.

For the displayed tableau,

$$Ac = \left(0, 0, \frac{1}{32}, \frac{8}{49}, \frac{49}{162}, 0, \frac{1}{2}, \frac{8}{49}\right)^\top,$$

so that  $(Ac)_i = \frac{1}{2}c_i^2$  at the six stages  $i = 1, 3, 4, 5, 7, 8$ , whereas the two remaining stages both have  $c_i = \frac{1}{5}$  and  $(Ac)_i = 0$ . Hence

$$q_1 = Ac - \frac{1}{2}c^{\odot 2} = \left(0, -\frac{1}{50}, 0, 0, 0, -\frac{1}{50}, 0, 0\right)^\top.$$

Consequently,

$$\widehat{b} \odot q_1 = \left(0, \frac{1}{250}, 0, 0, 0, -\frac{1}{250}, 0, 0\right)^\top \neq 0.$$

At  $p = 5$ , every admissible pair  $(m, n)$  has  $m \geq 2$ , so  $q_1 \in Q_m$  and  $QO(m)$  fails for every such pair. The method has order five nevertheless, disproving the conjectured necessity.  $\square$

Scope. For the same formula,

$$\widehat{b}^\top q_1 = 0.$$

More generally, the third-order conditions imply

$$b^\top q_1 = b^\top Ac - \frac{1}{2}b^\top c^{\odot 2} = \frac{1}{6} - \frac{1}{2} \cdot \frac{1}{3} = 0.$$

Thus the result concerns the coordinatewise condition  $b \odot Q_m = \{0\}$  as printed and does not refute a modified conjecture based on scalar orthogonality. The sufficiency theorem itself is unaffected.

## REFERENCES

- [1] J. He and J. Huang, *Low stage high order explicit Runge–Kutta methods via Q- and D-conditions: general theory and efficient recursive construction*, arXiv:2605.16995v1, 2026.
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